

Sarthak Shah

Portfolio | LinkedIn | Github | notesarthakshah@gmail.com | X

SKILLS

Languages: Rust, TypeScript, Go, Solidity, Lua

Solana: Anchor, Pinocchio, Codama, Carbon, Yellowstone, Jetstreamer

EXPERIENCE

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| Founding Engineer | MetEngine | June 2025 – Present |
| <ul style="list-style-type: none">• Researched, Designed and Implemented protocols across AMMs, Liquidity Providing, Lending & Borrowing, Yield-bearing Vaults, Perps, and Launchpads• Owned the LP Trading Terminal (\$114m in volume) end to end: parsing (manual, codama, carbon), indexing (yellowstone grpc), backfills (jetstreamer firehose), backend (axum), transport (kafka), storage (clickhouse, postgres), and infra + deployments + observability (aws, ci/cd)• Shipped the Chrome Extension: 2.2k users in 3 weeks, 90 paid at 1 SOL | | |
| Software Development Engineer | Gasyard Finance | December 2024 - May 2025 |
| <ul style="list-style-type: none">• Contributed to Intent Based Cross-Chain Bridge routes (Solana, EVM and MoveVM) for EVM/Non-EVM networks across Testnet/Mainnet environments that did \$550K in Volume and 120K+ requests in 5 months.• Developed the Fastify Backend and a high-performant Gasyard Bridge Explorer• Created Fuelbae, an ai agent executing defi actions across Aave, Uniswap, Hyperliquid, and Curve. | | |
| Software Engineer | Orbit (acq. Forward Research) | March 2024 - December 2024 |
| <ul style="list-style-type: none">• One of the first 10 builders on AO, contributing during the ecosystem's earliest days.• Contributed to the Lua smart contracts powering Orbit's oracle, the first protocol deployed on AO, enabling price feeds, data feeds, and arbitrary API access on Arweave's actor-oriented compute layer.• Built and scaled the developer community from 0 to 2,000+ members, providing direct technical support to 10+ integration partners.• Authored 4 developer blogs, comprehensive documentation, guides, and working demo applications that accelerated adoption of Orbit and reduced developer support tickets by 40%.• Tested Orbit nodes before mainnet launch, identifying edge cases and stability issues that improved network reliability. | | |
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PROJECTS

- Gossip Toy** - Reimplementing Solana's Gossip Protocol from Scratch [Github](#)
- Built a complete peer-to-peer gossip network in **Rust** over UDP, implementing Solana's pull/push protocol, CRDS replication, liveness detection (Ping/Pong), and binary message serialization.
 - Implemented **Bloom filter-based anti-entropy synchronization** and timestamp-based conflict resolution, enabling efficient state reconciliation while minimizing redundant network traffic.
 - Designed a multi-threaded peer-to-peer system where nodes converge to a consistent replicated state using CRDS, Bloom filter anti-entropy synchronization, and UDP-based pull/push gossip.
- Nock** - Per-maker Onchain Order Book Exchange (CLOB) [Github](#)
- Architected a **per-maker order book** on Solana where each market maker owns an independent PDA, eliminating write contention and enabling concurrent quote updates under Sealevel's parallel execution model.
 - Implemented an **on-chain quote freshness mechanism** that automatically expires stale quotes using slot-based liveness checks without requiring external keepers.
 - Built a **native Pinocchio-based Solana program** with deterministic fixed-size account layouts and optimized binary size for low compute overhead.
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FELLOWSHIPS & HONORS

- **Contributed to OSS (Agave, Prisma, Quasar, SocketIO, and many more)**
 - Worked extensively using multiple Solana Protocols.
 - Won ~20 Hackathons.
 - **Member at Superteam | Based Builder of the Year - Farcaster Builders International (FBI)**
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EDUCATION

B.Tech. in Computer Science and Engineering, ITMBU University
(2021 - 2025)

CGPA: 8.5